

Coral Reefs

A stylized, monochromatic blue-toned illustration of an underwater scene. The background is a deep navy blue. In the upper left, there are wavy, lighter blue shapes representing water currents or waves. Several small, white bubbles of varying sizes float in the upper left and upper right areas. In the center-left, a school of small, dark blue fish swims towards the right. In the lower right, a larger school of similar fish swims towards the left. The bottom of the image features stylized coral reefs and sea anemones in a lighter shade of blue. On the far left, a vertical, leafy branch-like structure extends from the bottom. The overall aesthetic is clean and modern.



Hi I'm Alex!

This project is part of my PhD research and I'm so appreciative for your participation!

What are we going to do?

01

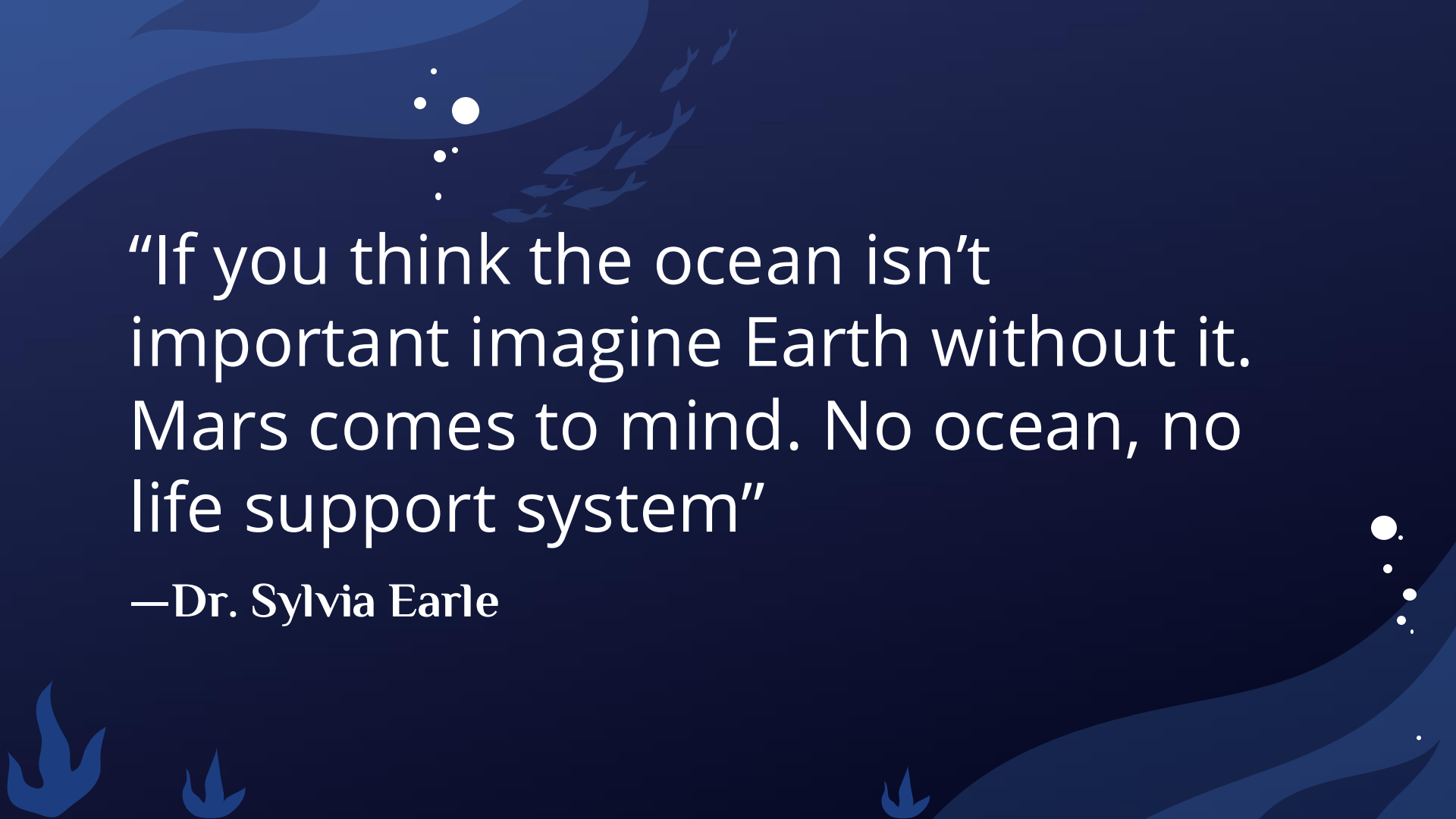
Introduction to
Coral Reef
Ecosystems

02

Virtual Field
Trip to Flower
Garden Banks &
Hawaii

03

Ocean Acidification
& Coral Growth Lab

The background is a dark blue gradient. In the upper left, there are several white dots of varying sizes, some arranged in a vertical line. In the upper right, there are stylized, flame-like or leaf-like shapes in a lighter blue color. In the lower left, there are more flame-like shapes. In the lower right, there are more white dots of varying sizes, some arranged in a vertical line.

“If you think the ocean isn’t
important imagine Earth without it.
Mars comes to mind. No ocean, no
life support system”

—Dr. Sylvia Earle



What do HUMANS need to survive?

FOOD

WATER

AIR

SHELTER

SLEEP

LOVE

WHAT IS A CORAL ...



A CORAL IS 3 THINGS!

ANIMAL

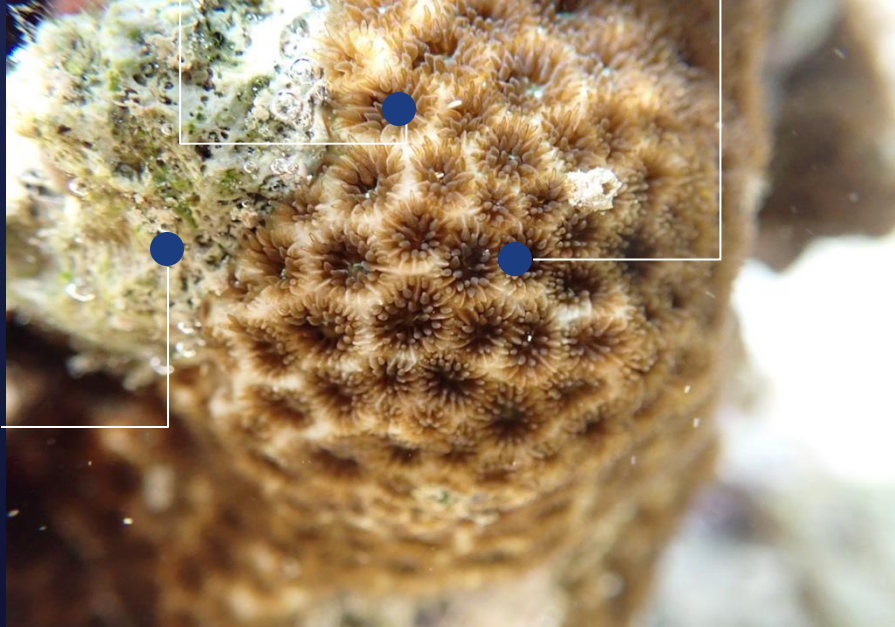
A coral is an animal in the phylum Cnidarian!

ROCK

Make a calcium carbonate limestone skeleton that builds a reef

PLANT

Symbiotic relationship with algae that live in the tissue – gives the coral it's color!





What do CORALS need to survive?

LIGHT/DEPTH

SALTWATER

WARMTH

PH

CLEAR WATER

PROTECTION



(ALGAE)

ZOOXANTHELLAE

CORAL CELL

They also give corals their beautiful colors.



The background is a dark blue gradient with stylized, lighter blue wavy lines representing water currents. In the upper right, a school of small fish swims towards the right, with several white bubbles of varying sizes nearby. In the lower left, another school of fish swims towards the left, also accompanied by white bubbles.

01

Coral's are animals



Corals are Cnidarians!

Similar to jellyfish!

- Colonial & sedentary
- Polyp = unit of a coral
- Individual polyps connected by tissue
- Polyps bud clones (asexual reproduction)



Types of Corals

- **Colonial**

- Warm, shallow waters
- Reef builders
- Photosynthesize
- Multiple polyps



- **Solitary**

- Not restricted to warm shallow waters
- Don't need to photosynthesize
- Single polyp
- Non-reef building



Heterotrophic feeding – catching tiny floating animals called zooplankton

- Mostly done at night
- Prey are pulled into the polyps mouth and digested by the stomach
- During the day = autotrophic feeding



Huffmyer, A., 2017

They can eat 2 different ways!

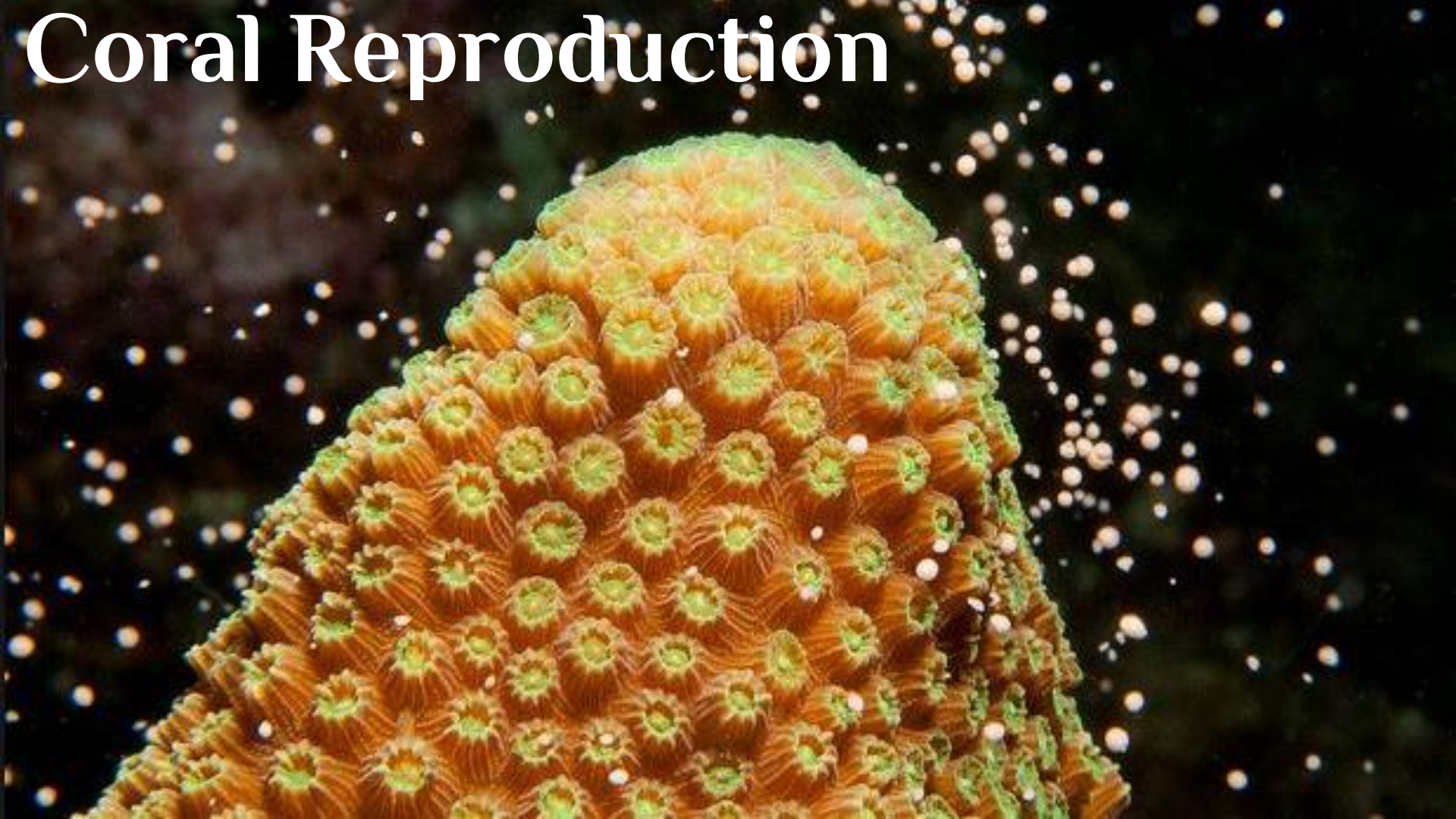
Heterotrophic
(Filter Feeding)



Autotrophic
(Photosynthesis)



Coral Reproduction



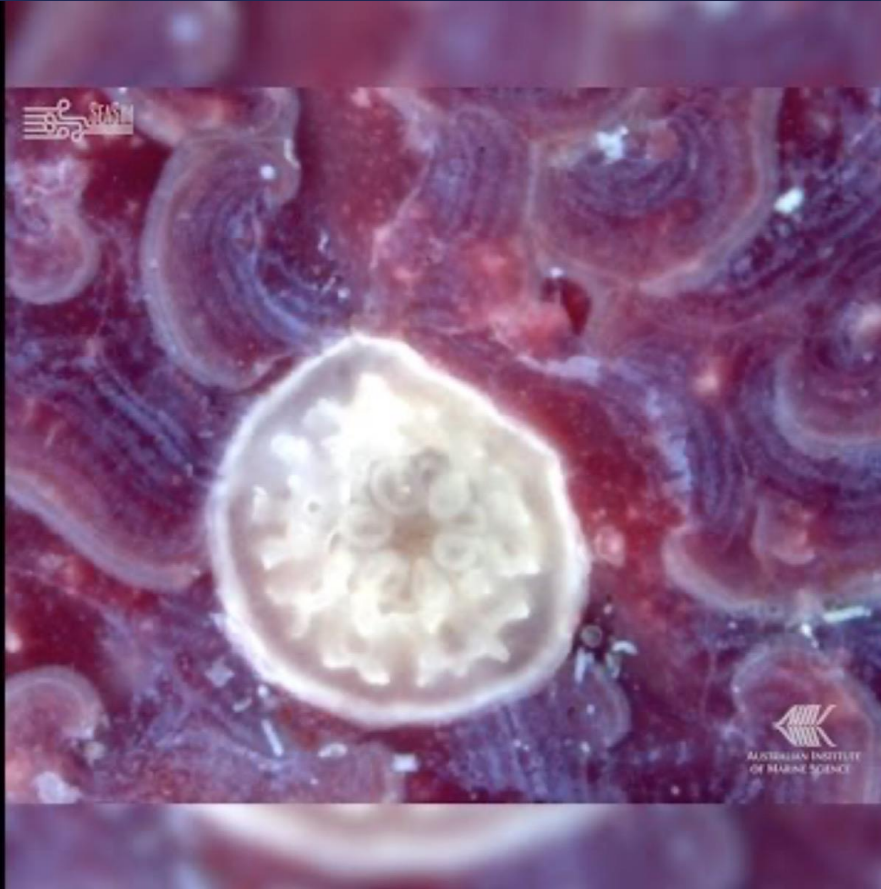
NATURE



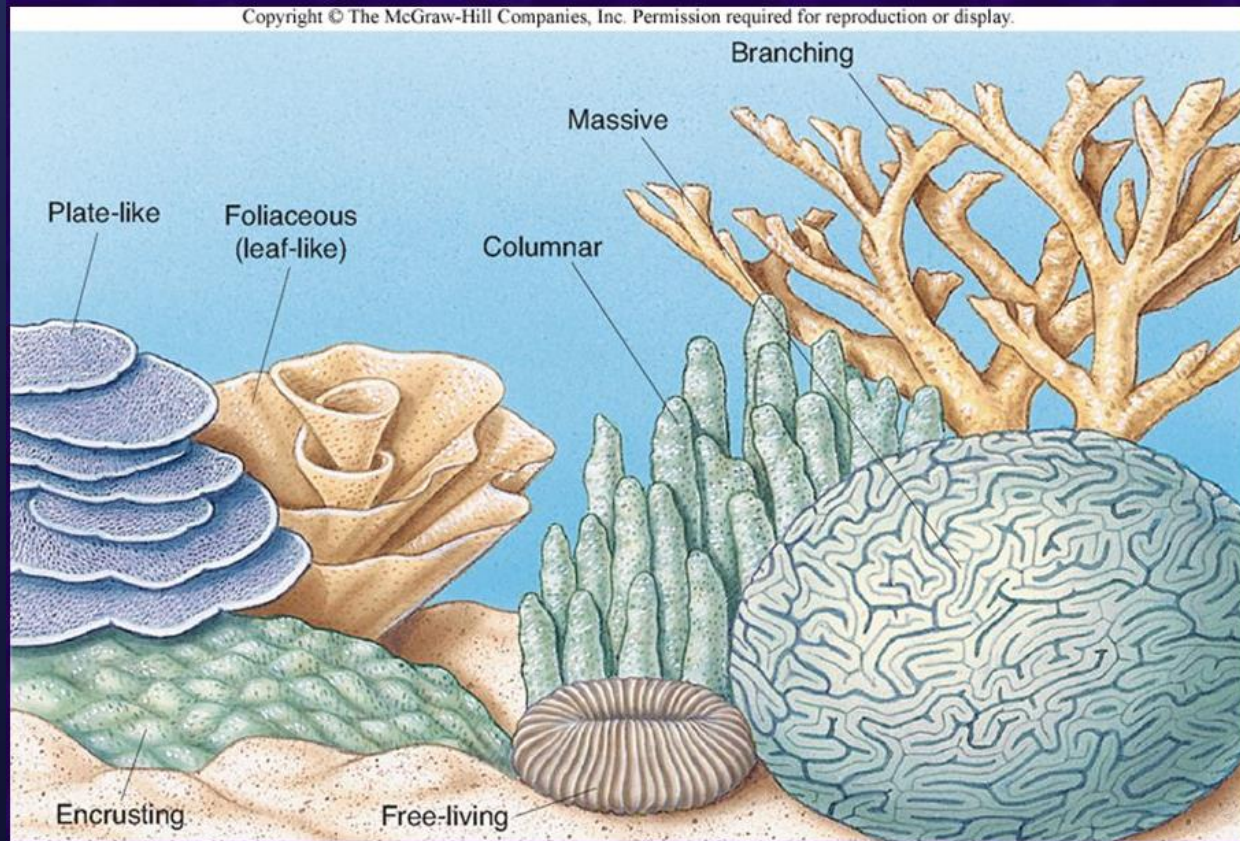
Witness a Massive Coral Spawning



Larvae settlement



Coral Colony Growth Forms



*You must be able to label this for your exams

The background is a dark blue gradient with stylized, lighter blue wavy lines representing water currents. In the upper right and lower left corners, there are clusters of small, light blue fish swimming in the same direction. Several white circles of varying sizes are scattered throughout, representing bubbles.

02

Coral's are plants

Symbiosis + - or 0

“The living together of unlike organisms”- Anton de Bary (1878)

Species

Commensalism

Competition

Parasitism

Predation

Mutualism

A

B





Coral Competition

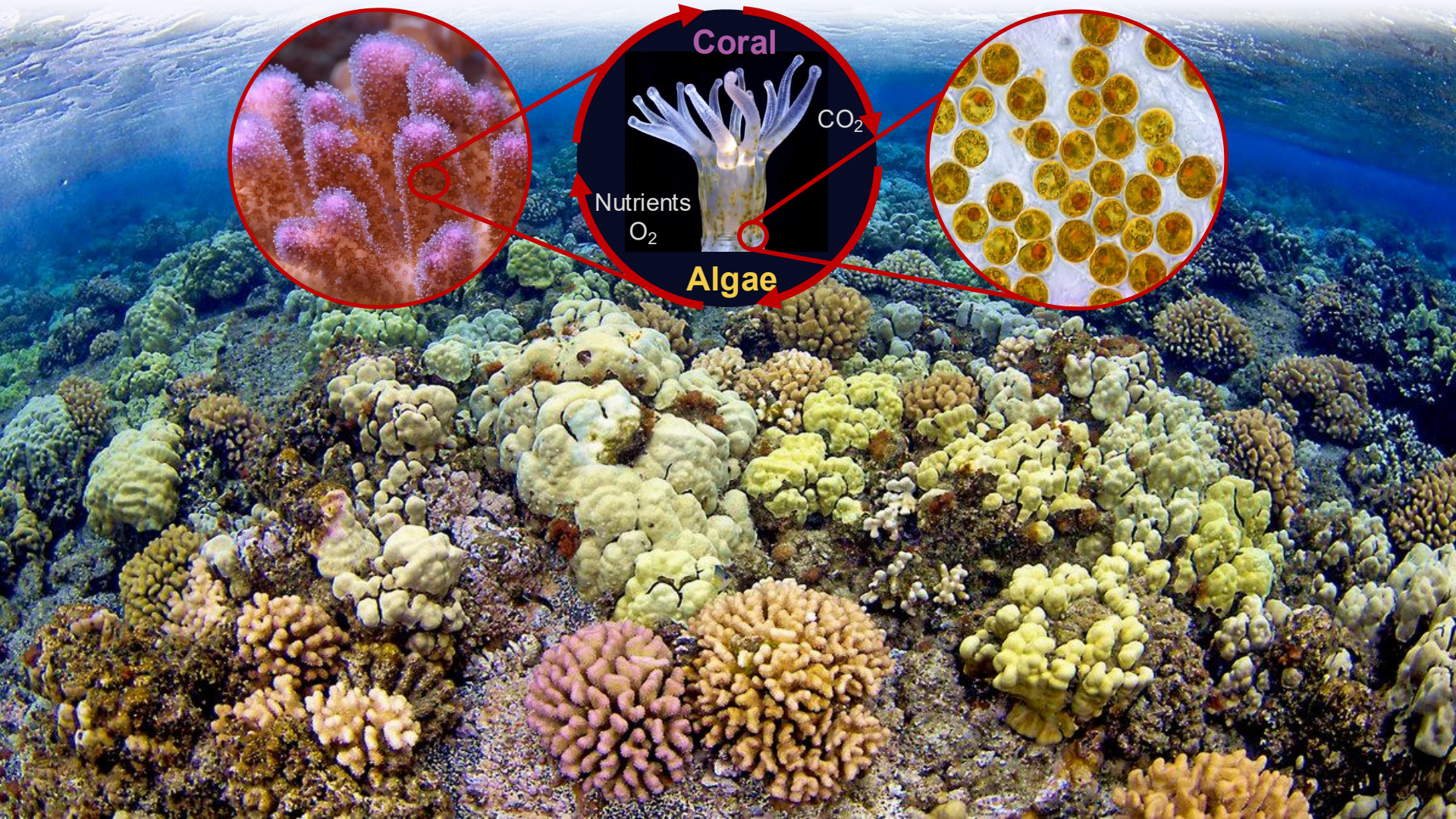


Competition

Direct = sweeper tentacles and nematocysts (sting!)

Indirect = overgrowing





Coral

CO₂

Nutrients

O₂

Algae

Corals and their symbionts

- Zooxanthellae = Symbiotic algae living in coral tissue
- Obligate, nutrient-sharing mutualism
- Millions per cm^2
- Photosynthesis (autotrophic feeding) provides the coral with over 90% of the energy it needs to survive



Benefits and Costs of Coral Symbioses



Benefits	Costs
Animal	
Supply of reduced carbon	Regulation of algal growth
Increased growth and repro	Defense against high oxygen
Increased calcification rate	Mechanisms for rejection of foreign or excess algae
Conservation of nutrients	Vulnerability to environmental stresses that affect plants
Zooxanthellae	
CO ₂ and nutrients from host	Translocation of a significant fraction of photosynthate to animal
Maintenance in photic zone	Regulation of growth rate, rate slower in coral than in free-living state
Protection from UV damage by animal tissues	Expulsion from host
Maintenance under uniform environmental conditions	Supply of CO ₂ and nutrients limited by host

Coral Bleaching

Corals still
have algae
in their
tissue



Corals are white
because they
have lost their
algae

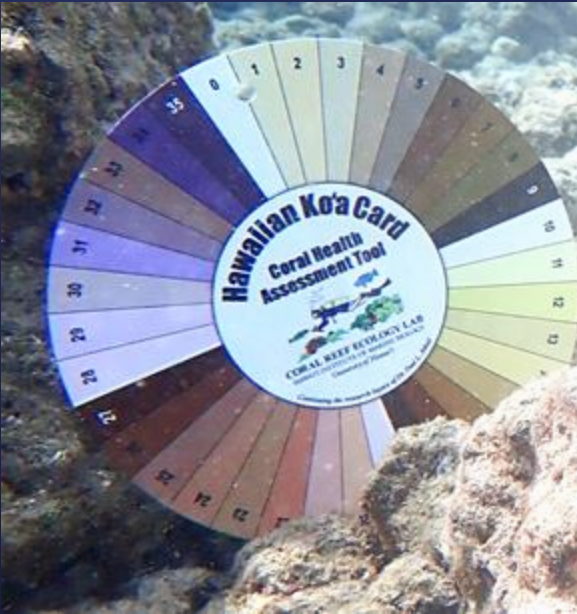


Healthy VS Bleached

Catlin seaview survey

The Hawaiian Ko'a Card

a coral health assessment tool



CORAL REEF ECOLOGY LAB
HAWAII INSTITUTE OF MARINE BIOLOGY
University of Hawai'i



How to use the Ko'a Card: Let's Survey!

Survey



Coral Shape



Coral Color



Report

Hawaiian Ko'a Card: Coral Health Assessment Tool



DATA ENTRY FORM

Coral Color 

Coral Shape 

Coral Species 

show scientific names

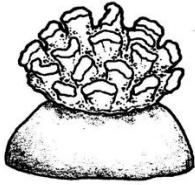
Coral Bleaching: 

Depth: 

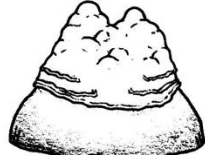
Water Temperature: 

Coral Shape

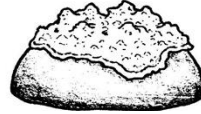
BRANCH



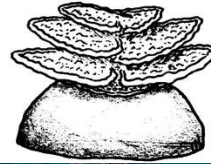
MOUND



CRUST

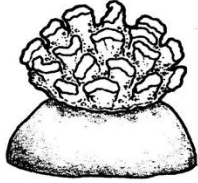


PLATE

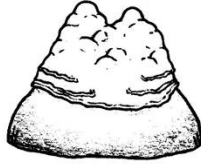


What shape is this color?

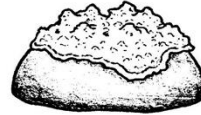
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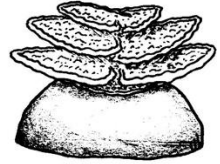
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CRUST

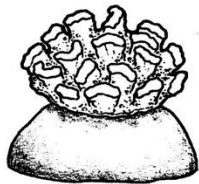


PLATE

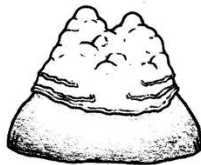


What shape is this color?

BRANCH



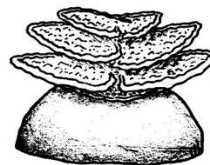
MOUND



CRUST



PLATE



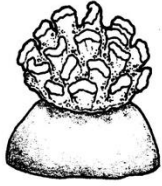
Select color that best represents the whole coral colony!



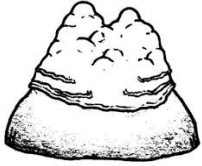
Select color that best represents the whole coral colony!



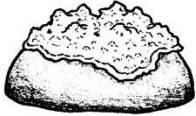
BRANCH



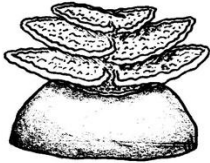
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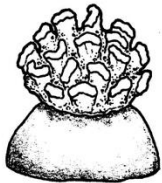
CRUST



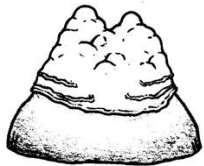
PLATE



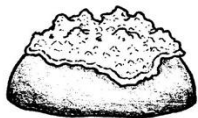
BRANCH



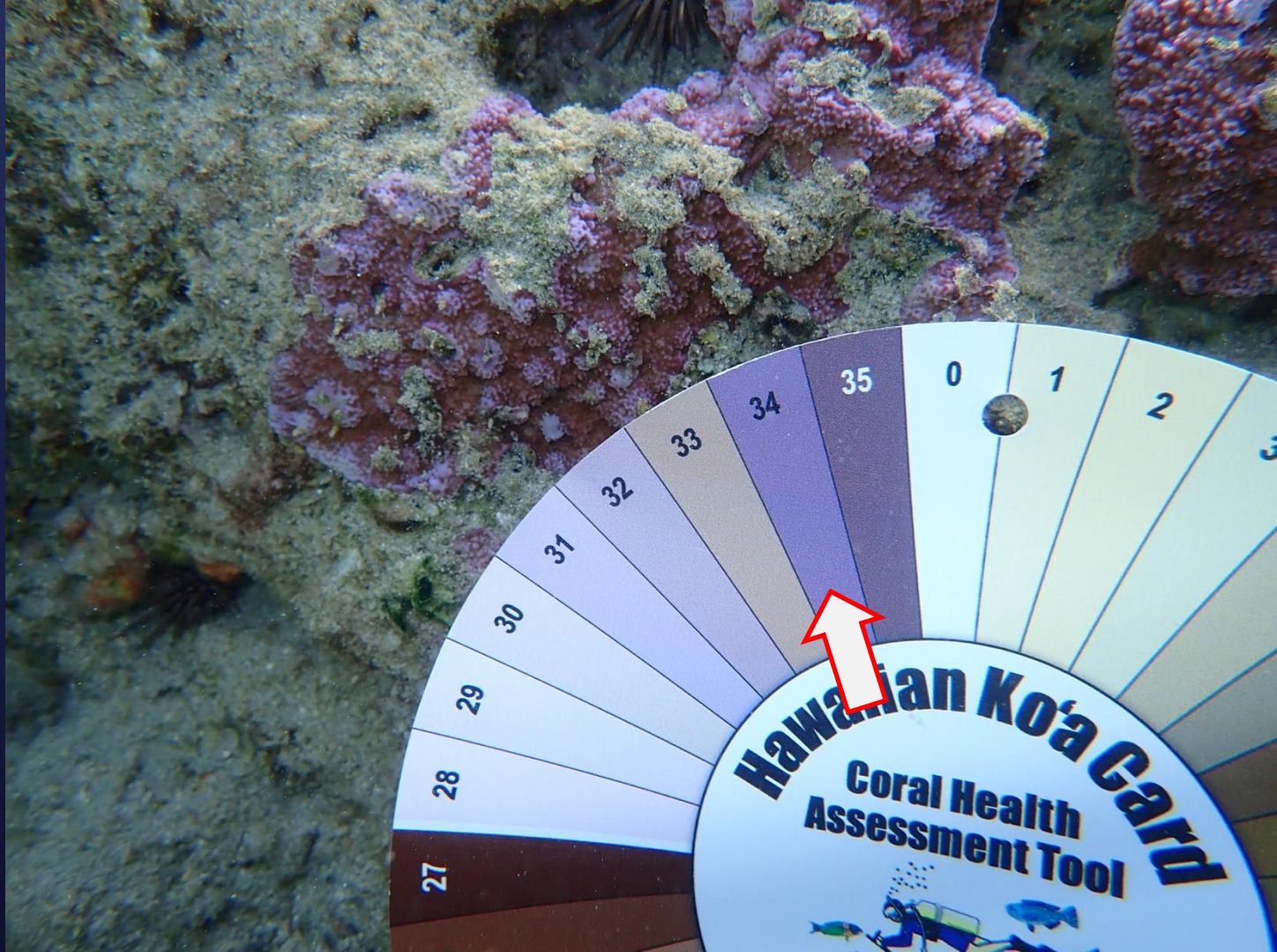
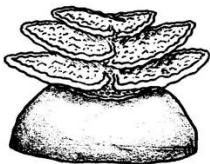
MOUND



CRUST



PLATE





01

Coral's are rocks

What is a reef?

- A reef = hard bottom structure composed by organisms and used by other organisms
 - Reefs themselves are not alive
- A **Coral Reef** is a reef build from the carbonate skeletons of corals



Oyster reefs



Tubeworm reefs



Calcareous algal reefs

Coral Growth (Calcification)

- Calcification = Process by which corals produce calcium carbonate (CaCO_3) skeleton
- 30% of corals' energy budget is devoted to calcification
- Calcification is accelerated by light = photosynthesis-driven process
- Increases surface area & structural support for the organism
- A way of maintaining a close proximity to high light levels
- Means for keeping up with sea level rise

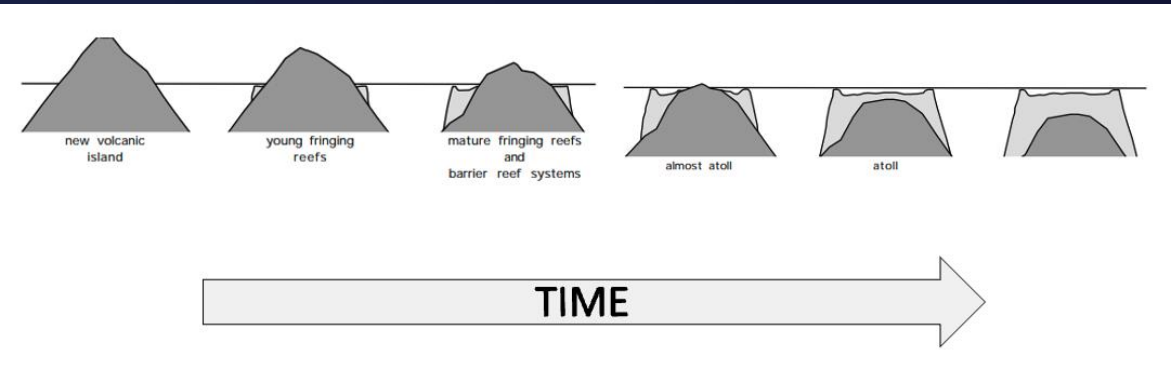


Coral Reef Geological Formation

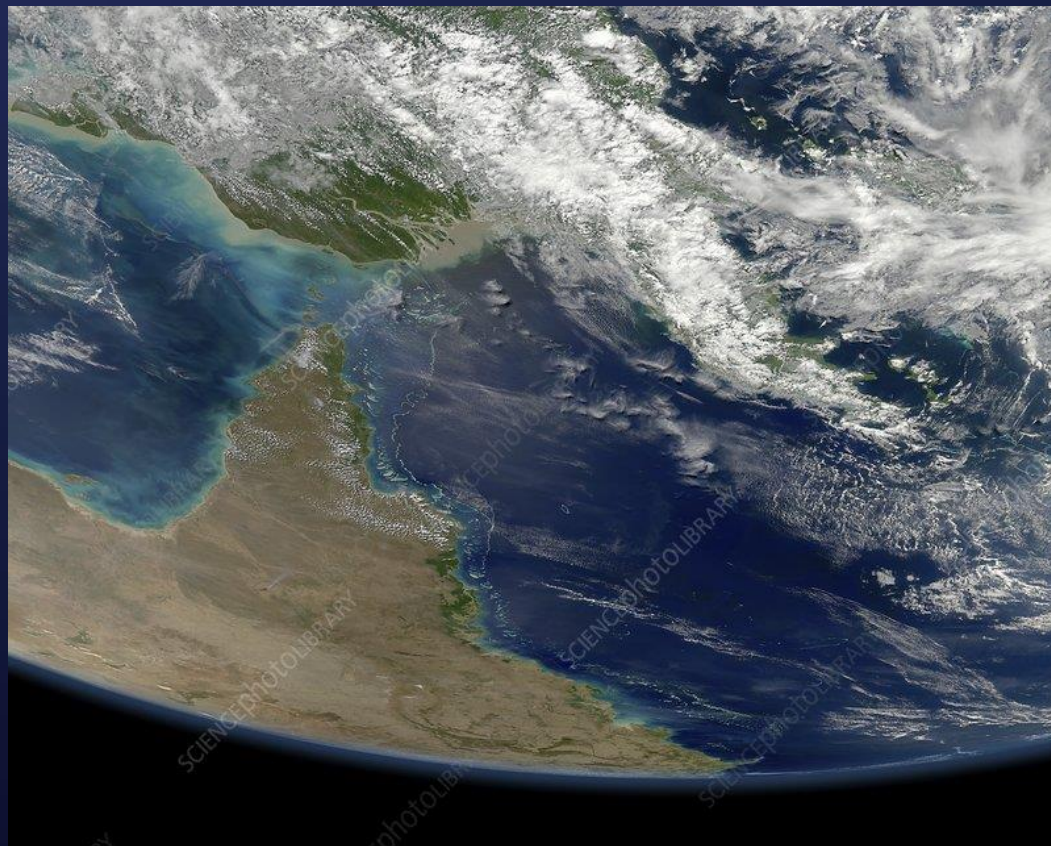
The biology makes the geology, that geology then supports the next generation of biology

- Coral reefs are one of the older reef systems on Earth
- Coral reefs are landforms that evolve through the growth of living organisms (coral)
- Skeletons of calcium carbonate that accumulate into limestone deposits

Darwin wrote "The Structure and Distribution of Coral Reefs" in 1842, 17 years before his better-known "The Origin of Species".

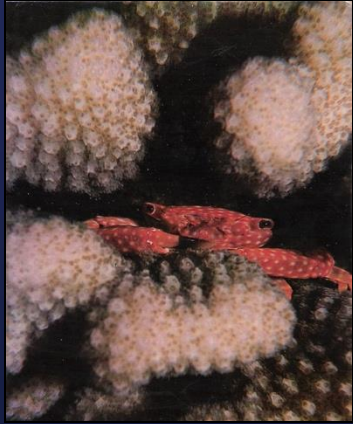


Corals are massive - GBR



Symbioses on the Reef

Trapezoid crabs & Coral



Clownfish & Anemone



Cleaner wrasse



Herbivores & Turtles



Gobies & Shrimp

